TECHNOFuture
KAFR EL SHEIKH

Afro-Asian Forum for Innovation and Technology
AFFIAT · Aswan Finals 2026

MEDICATION CARE · EGYPT

MedSync

Never miss a dose.

Doctors prescribe, patients get an alarm at dose time, and nurses mark every dose, so everyone can see who is on track.

01 · WHY Medicine only works if you take it.

THE PROBLEM المشكلة

Doses get forgotten.

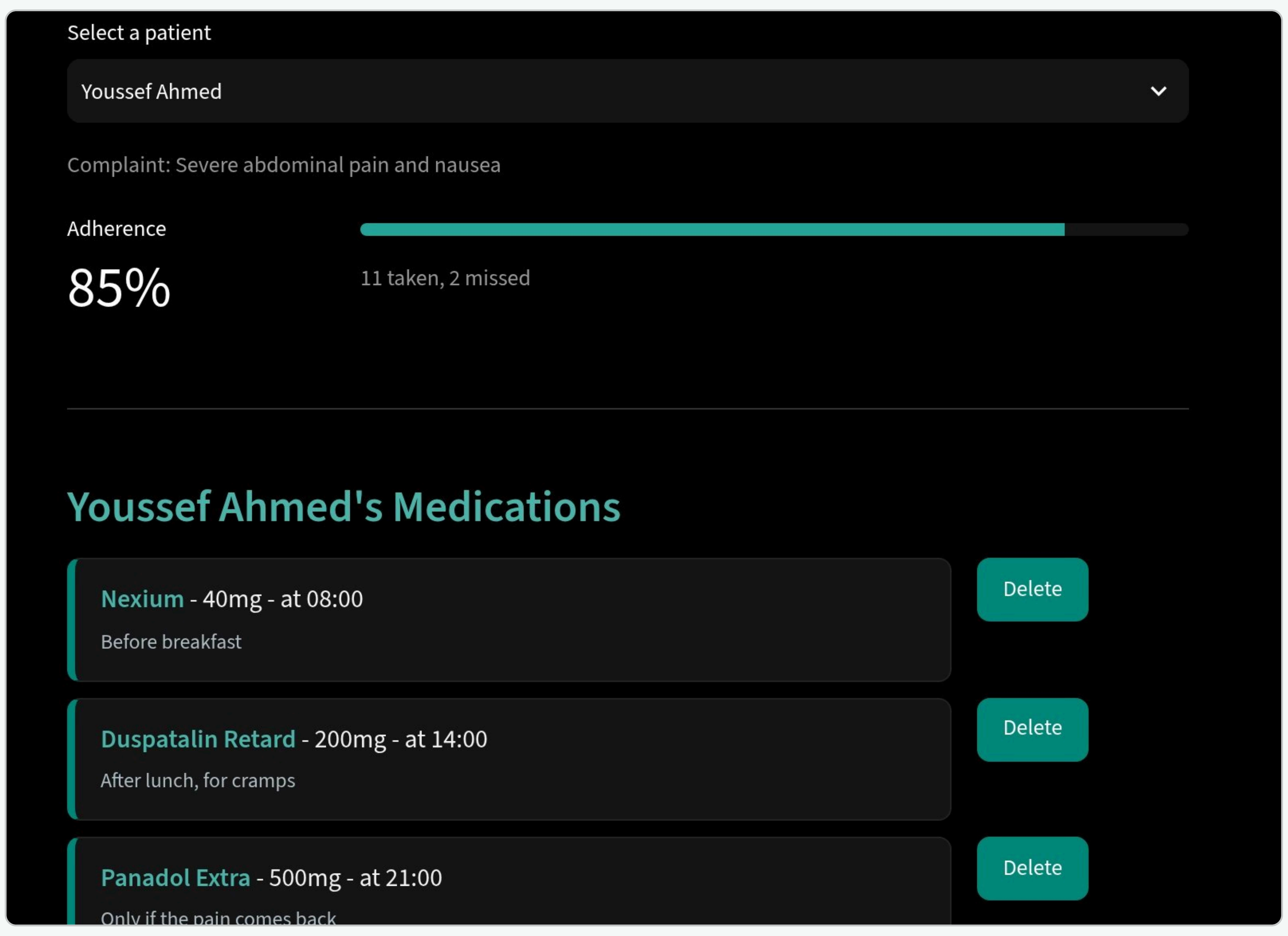
Patients who take medicine every day miss a dose or take it late. The doctor usually finds out only at the next visit, when the patient is already worse.

OUR SOLUTION الحل

Remind, record, review.

MedSync rings an alarm at each dose time, lets the nurse mark every dose, and shows the doctor and the patient an adherence score.

02 · SEE IT From prescription to dose. من الوصفة إلى الجرعة



Select a patient

Youssef Ahmed

Complaint: Severe abdominal pain and nausea

Adherence

85%

11 taken, 2 missed

Youssef Ahmed's Medications

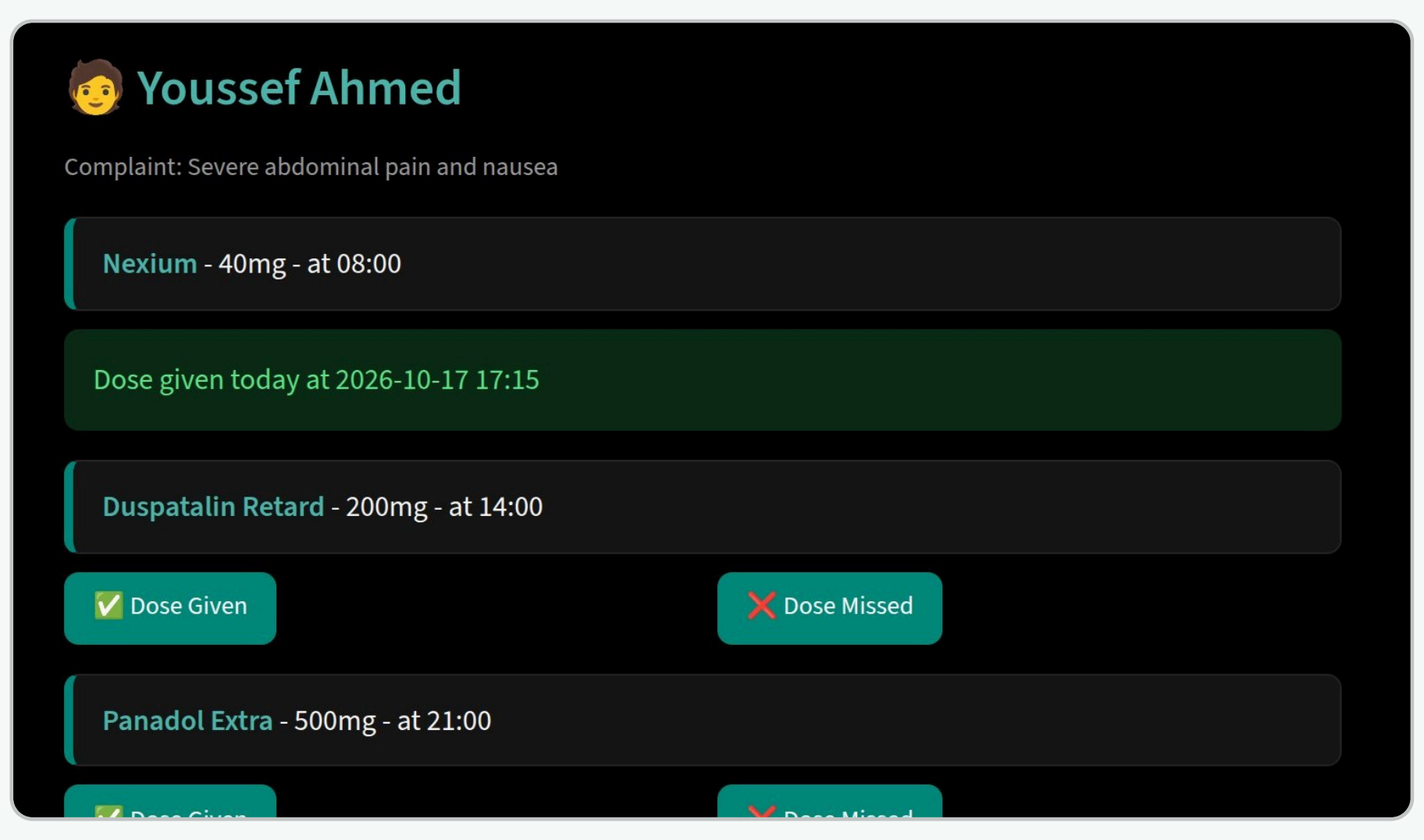
Nexium - 40mg - at 08:00
Before breakfast

Duspatalin Retard - 200mg - at 14:00
After lunch, for cramps

Panadol Extra - 500mg - at 21:00
Only if the pain comes back

The doctor's "My Patients" tab: Youssef's complaint, his adherence (11 of 13 doses, 85%), and his three medicines with their dose times. Below them, the form to prescribe from 207 medicines.

- **Prescribe in seconds**
  Search 207 medicines by name, then set the dosage, the dose time and a note.
- **A care team for each patient**
  New patients wait in a list until a doctor assigns them a nurse.
- **An alarm at dose time**
  The patient's page shows a banner, beeps and sends a browser notification.
- **Every dose is marked**
  The nurse taps "Dose Given" or "Dose Missed", once per medicine each day.
- **Adherence at a glance**
  The share of doses taken, for the doctor and the patient, with a full history.
- **Visit notes and a chatbot**
  Doctors log diagnoses. The chatbot answers from the database, and asks Gemini the rest.



🧑 Youssef Ahmed

Complaint: Severe abdominal pain and nausea

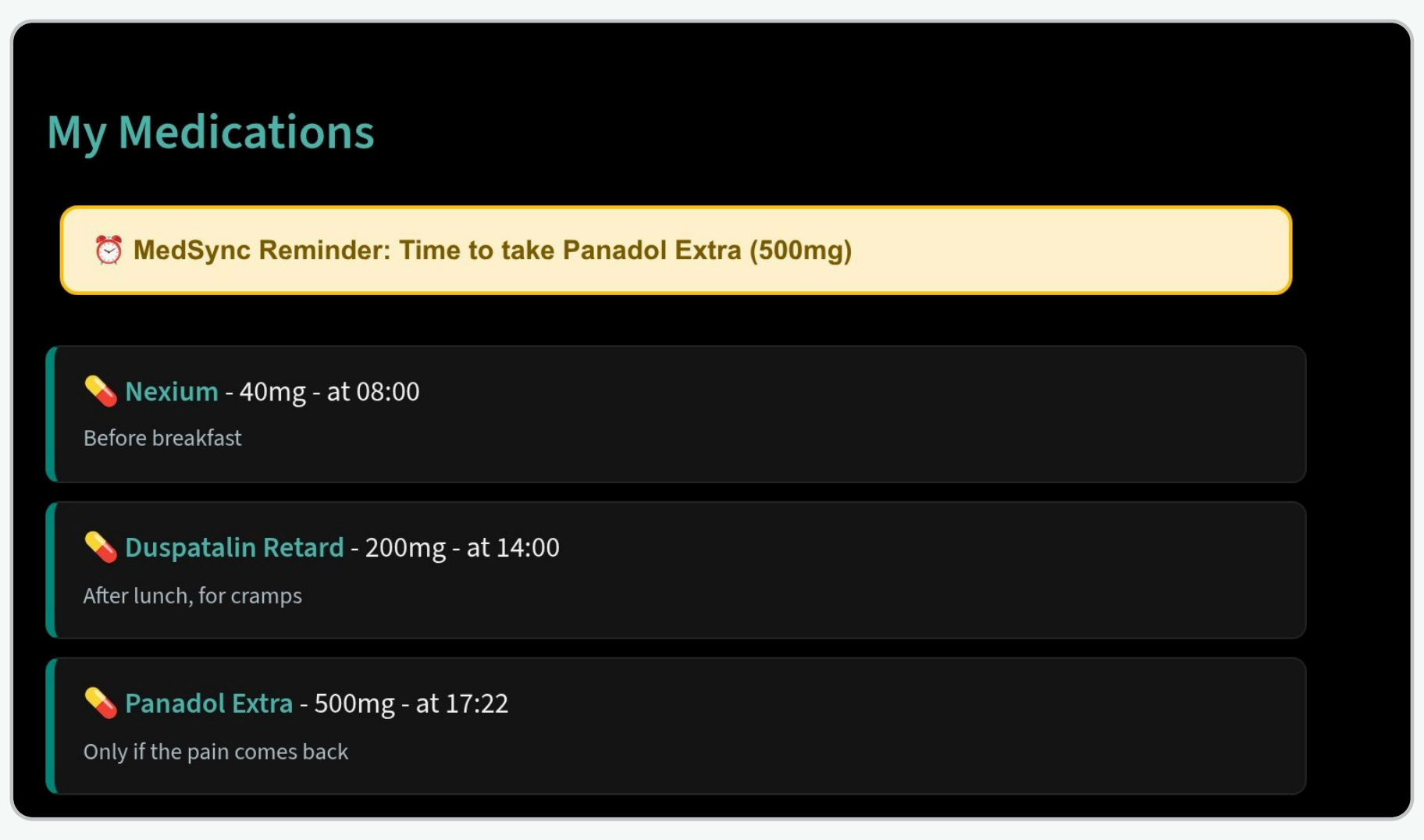
Nexium - 40mg - at 08:00

Dose given today at 2026-10-17 17:15

Duspatalin Retard - 200mg - at 14:00

Panadol Extra - 500mg - at 21:00

Nurse: today's doses, given or missed.



My Medications

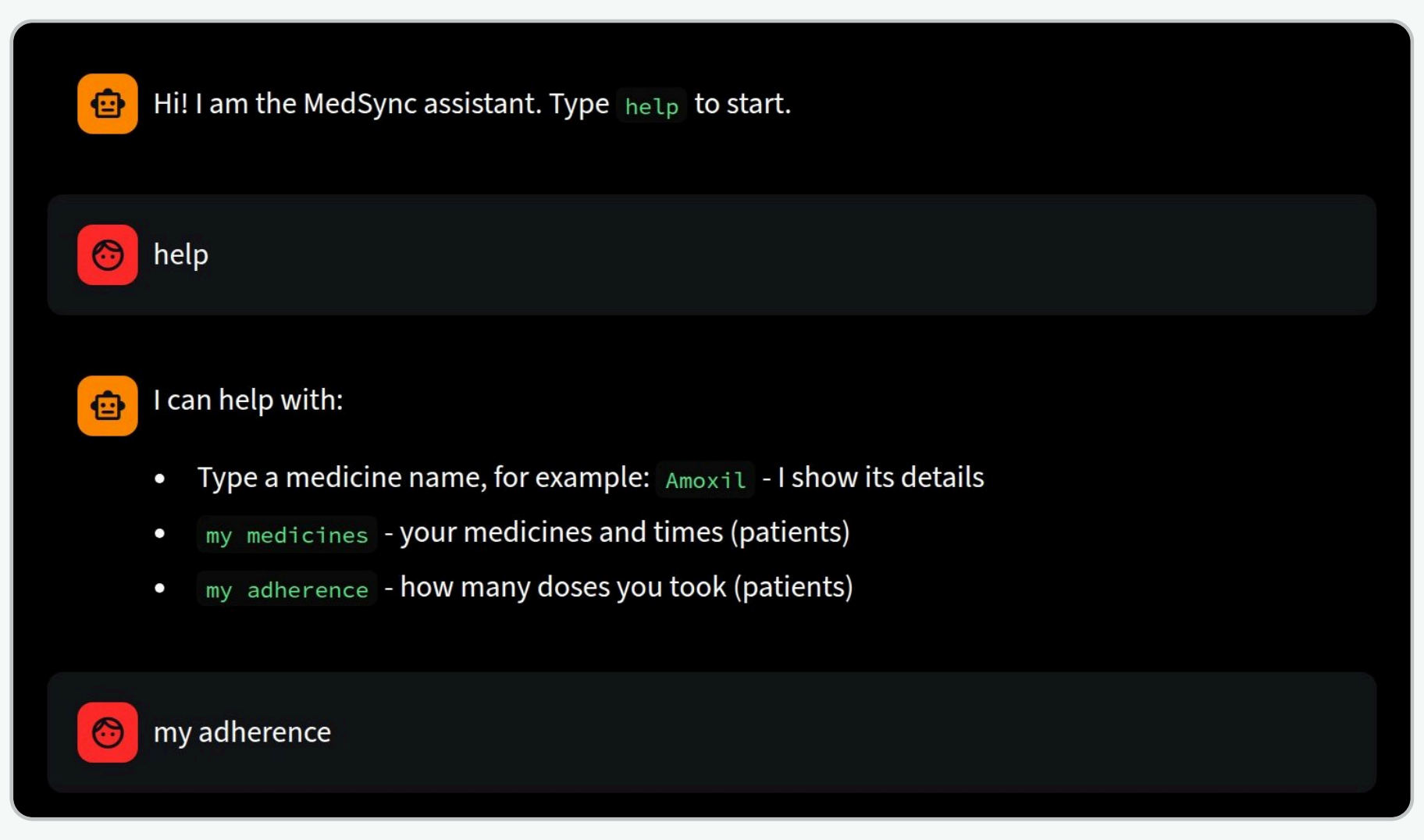
⏰ MedSync Reminder: Time to take Panadol Extra (500mg)

Nexium - 40mg - at 08:00
Before breakfast

Duspatalin Retard - 200mg - at 14:00
After lunch, for cramps

Panadol Extra - 500mg - at 17:22
Only if the pain comes back

Patient: the alarm banner at dose time.



Hi! I am the MedSync assistant. Type `help` to start.

help

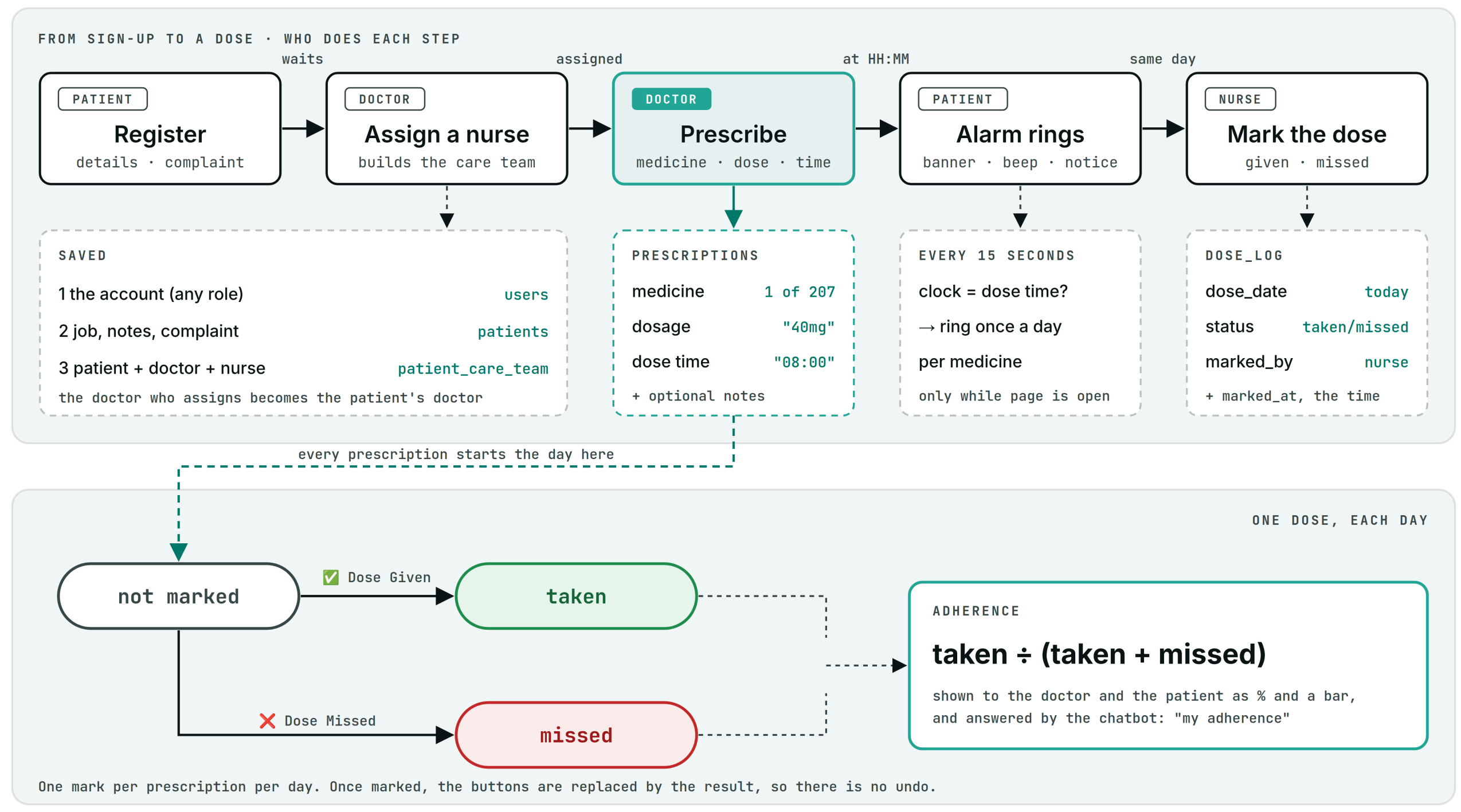
I can help with:
- Type a medicine name, for example: `Amoxil` - I show its details
- `my medicines` - your medicines and times (patients)
- `my adherence` - how many doses you took (patients)

my adherence

Chatbot: "help" and "my adherence", no internet needed.

03 · HOW Plain Python, no JOIN. بايثون بسيط بدون JOIN

FROM SIGN-UP TO A DOSE



FROM SIGN-UP TO A DOSE · WHO DOES EACH STEP

SAVED
1 the account (any role) — users
2 job, notes, complaint — patients
3 patient + doctor + nurse — patient_care_team
the doctor who assigns becomes the patient's doctor

PRESCRIPTIONS
medicine — 1 of 207
dosage — "40mg"
dose time — "08:00"
+ optional notes

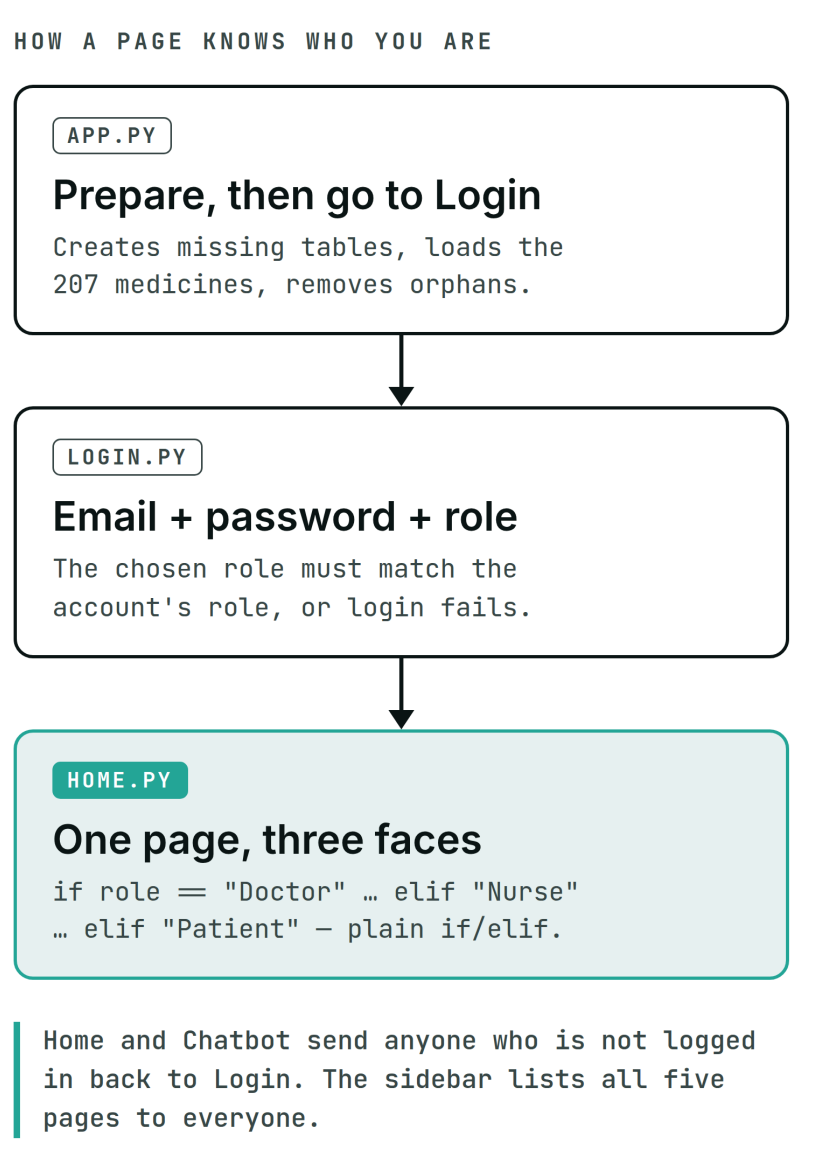
EVERY 15 SECONDS
clock = dose time?
→ ring once a day
per medicine
only while page is open

DOSE_LOG
dose_date — today
status — taken/missed
marked_by — nurse
+ marked_at, the time

ADHERENCE
taken ÷ (taken + missed)
shown to the doctor and the patient as % and a bar, and answered by the chatbot: "my adherence"

One mark per prescription per day. Once marked, the buttons are replaced by the result, so there is no undo.

WHO SEES WHAT



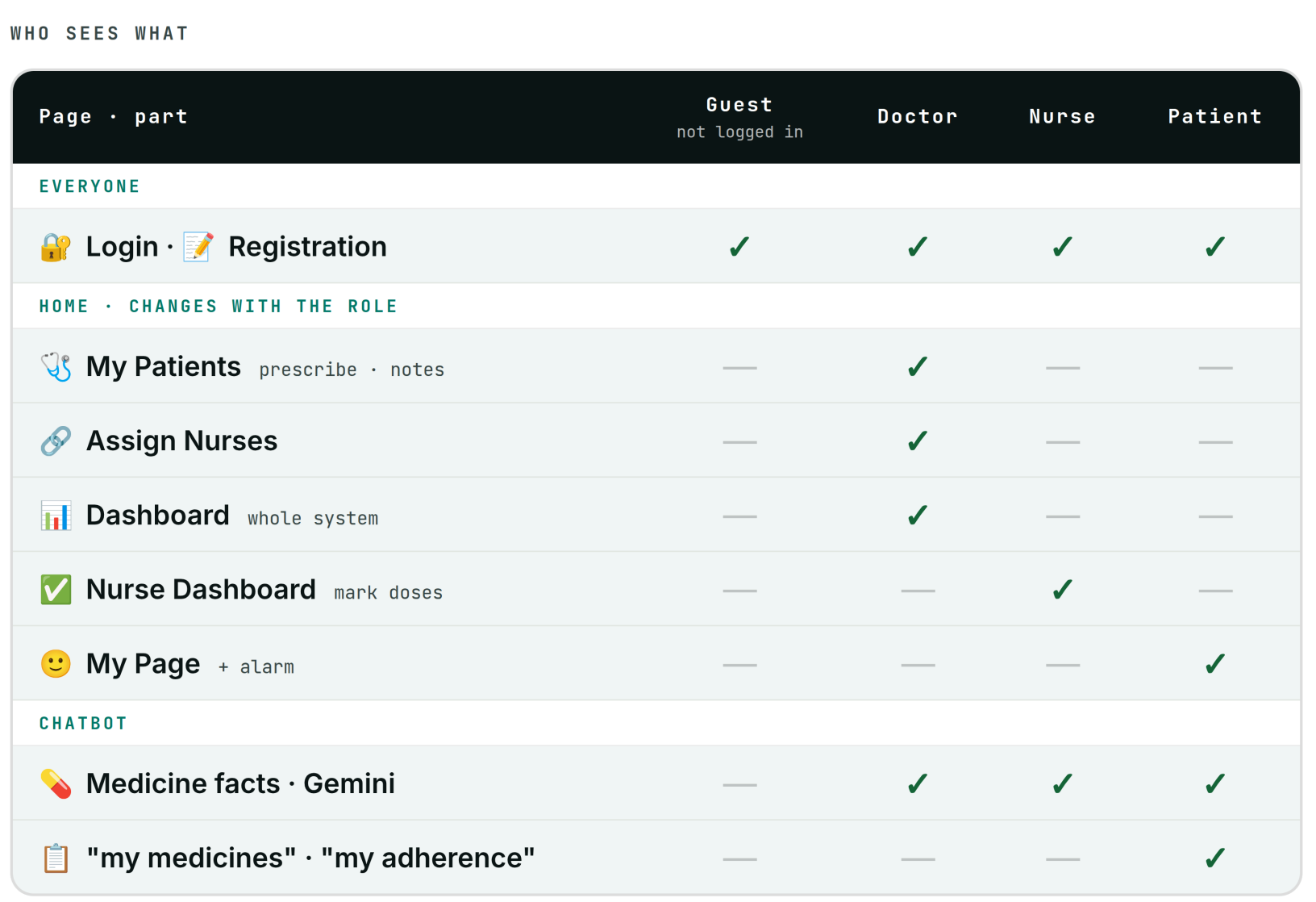
HOW A PAGE KNOWS WHO YOU ARE

Prepare, then go to Login
Creates missing tables, loads the 207 medicines, removes orphans.

Email + password + role
The chosen role must match the account's role, or login fails.

One page, three faces
if role == "Doctor" … elif "Nurse" … elif "Patient" – plain if/elif.

Home and Chatbot send anyone who is not logged in back to Login. The sidebar lists all five pages to everyone.



WHO SEES WHAT

Page · part	Guest not logged in	Doctor	Nurse	Patient
EVERYONE				
🔐 Login · 📝 Registration	✓	✓	✓	✓
HOME · CHANGES WITH THE ROLE				
🩺 My Patients prescribe · notes	—	✓	—	—
🔗 Assign Nurses	—	✓	—	—
📊 Dashboard whole system	—	✓	—	—
✅ Nurse Dashboard mark doses	—	—	✓	—
🙂 My Page + alarm	—	—	—	✓
CHATBOT				
💊 Medicine facts · Gemini	—	✓	✓	✓
📋 "my medicines" · "my adherence"	—	—	—	✓

3
roles, one login: doctor, nurse and patient

207
medicines loaded from medicines_seed.sql

9
SQLite tables, from users to dose_log

0
SQL JOINs: tables are linked with for loops

BUILT WITH

`Python` `Streamlit` `SQLite` `Gemini API` `HTML + CSS + JS`

Sief & Leen

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A school demo with made-up patients. Not for real medical use.

Live demo
at our table